

Deep Learning for Natural Language Processing

Assignment 6 — Solutions

Week 6: Pretraining & Transfer Learning

Total: 7 Questions | 10 Marks | 4×1 Mark, 3×2 Marks

1. A BERT encoder is pretrained with the masked language modelling objective. One training instance is the nine-token input

$[\text{CLS}]_1 \text{ the}_2 \text{ technician}_3 [\text{MASK}]_4 \text{ the}_5 \text{ faulty}_6 [\text{MASK}]_7 \text{ quickly}_8 [\text{SEP}]_9$

where subscripts denote positions. Which context is available when the model computes its output distribution at position 4, and how many loss terms does this instance contribute? **[1 Mark]**

- A) Positions 1–3 only; nine loss terms
- B) All other positions; nine loss terms
- C) Positions 1–3 only; two loss terms
- D) All other positions; two loss terms

Answer & Solution

Answer: D) All other positions; two loss terms

Solution:

Self-attention in an encoder carries no causal mask, so position 4 is contextualised by every other position on both sides, including the $[\text{MASK}]$ placeholder at position 7. The objective is $p_\theta(x \mid \tilde{x})$: a prediction is scored only where the original token was removed. Positions 4 and 7 are the masked ones, so the instance contributes two loss terms.

Therefore, the correct option is **D**.

Reference: Lecture 27, Slide 5 (PDF page 34) — Solution: Use Masks (Masked LM objective and the restriction of loss terms to masked positions)

2. A span-based model processes an input sequence of $T = 12$ tokens. A span runs from start i to end j with $1 \leq i \leq j \leq T$, and the application imposes a length limit $L = 4$, so that legal spans are those with $j - i < L$. How many legal spans does the model enumerate?

[1 Mark]

- A) 78
- B) 33
- C) 42
- D) 48

Answer & Solution

Answer: C) 42

Solution:

Step 1 — Convert the constraint to a length. $j - i < 4$ means $j - i \leq 3$, so a span has at most $(j - i) + 1 = 4$ tokens, i.e. $\ell \in \{1, 2, 3, 4\}$.

Step 2 — Count spans of each length. There are $T - \ell + 1$ spans of length ℓ :

$$\ell = 1 : 12, \quad \ell = 2 : 11, \quad \ell = 3 : 10, \quad \ell = 4 : 9.$$

Step 3 — Sum.

$$|S(x)| = 12 + 11 + 10 + 9 = 42.$$

Therefore, the correct option is **C**.

Reference: [Lecture 28, Slide 5 \(PDF page 63\) — Fine-tuning for span-based applications \(legal span set \$S\(x\)\$ under the length limit \$j - i < L\$ \)](#)

3. A BERT-based reading comprehension model uses the start vector $S = [1, 1]$ and end vector $E = [1, -1]$. The passage has six tokens whose final-layer representations are

$$T_1 = [6, 2], \quad T_2 = [2, 0], \quad T_3 = [3, -3], \quad T_4 = [3, -1], \quad T_5 = [5, -1], \quad T_6 = [3, -5].$$

The score of the candidate span from position i to position j is $S \cdot T_i + E \cdot T_j$, and a span is legal only if $j \geq i$ and $j - i < L$ with $L = 3$. Which span does the model predict, and what is its score? **[2 Marks]**

- A) Tokens 1 to 6, with score 16
- B) Tokens 1 to 1, with score 12
- C) Tokens 1 to 3, with score 14
- D) Tokens 5 to 6, with score 12

Answer & Solution

Answer: C) Tokens 1 to 3, with score 14

Solution:

Step 1 — Start scores $s_i = S \cdot T_i$.

$$s_1 = 8, \quad s_2 = 2, \quad s_3 = 0, \quad s_4 = 2, \quad s_5 = 4, \quad s_6 = -2.$$

Step 2 — End scores $e_j = E \cdot T_j$.

$$e_1 = 4, \quad e_2 = 2, \quad e_3 = 6, \quad e_4 = 4, \quad e_5 = 6, \quad e_6 = 8.$$

Step 3 — Apply the constraints. The best start ($i = 1$) and best end ($j = 6$) give $j - i = 5$, which is not $< L = 3$, so that pair is illegal. The search is restricted to $j \geq i$ with $j - i \leq 2$.

Step 4 — Maximise over the legal spans.

$i = 1:$	$(1, 1) = 12;$	$(1, 2) = 10;$	$(1, 3) = \mathbf{14}$
$i = 2:$	$(2, 2) = 4;$	$(2, 3) = 8;$	$(2, 4) = 6$
$i = 3:$	$(3, 3) = 6;$	$(3, 4) = 4;$	$(3, 5) = 6$
$i = 4:$	$(4, 4) = 6;$	$(4, 5) = 8;$	$(4, 6) = 10$
$i = 5:$	$(5, 5) = 10;$		$(5, 6) = 12$
$i = 6:$	$(6, 6) = 6$		

The maximum is 14, at the span (1, 3).
Therefore, the correct option is **C**.

Reference: Lecture 28, Slides 4–5 (PDF pages 62–63) — Fine-tuning for SQUAD (span score $S \cdot T_i + E \cdot T_j$ maximised subject to $j \geq i$) and the application-specific length limit $j - i < L$

4. A single T5 model is fine-tuned for two tasks: judging whether a sentence is grammatically acceptable (a two-way decision), and rating the similarity of a sentence pair on a continuous 1–5 scale. Under the text-to-text framework, how are the two outputs produced? **[1 Mark]**

- A) Both require task-specific output heads
- B) Only generation tasks are supported
- C) Classification as text; score by regression
- D) Both are generated as literal text

Answer & Solution

Answer: D) Both are generated as literal text

Solution:

In a text-to-text framework the input carries a task prefix and the target is the answer written as text. The acceptability label is emitted as the string `not acceptable`, and the similarity score as the characters `3.8` — generated token by token by the ordinary decoder, not read from a regression unit. Neither task adds an output head.

Therefore, the correct option is **D**.

Reference: Lecture 28, Slide 28 (PDF page 86) — T5 can be used for various tasks (text-to-text framework)

5. A decoder-only Transformer is pretrained as a language model on the five-token sequence $w_1 w_2 w_3 w_4 w_5$. After training, the model assigns

$$p(w_2 | w_1) = 0.5, \quad p(w_3 | w_1 w_2) = 0.25, \quad p(w_4 | w_1 w_2 w_3) = 0.5, \quad p(w_5 | w_1 \dots w_4) = 0.125,$$

and its unconditional distribution assigns probability 0.5 to w_1 . Using $\log_2 0.5 = -1$, $\log_2 0.25 = -2$ and $\log_2 0.125 = -3$, what is the total negative log-likelihood of the tokens that this pretraining objective actually predicts? **[2 Marks]**

- A) 8
- B) 7
- C) 1.75
- D) 3

Answer & Solution

Answer: B) 7

Solution:

Step 1 — Identify the predicted tokens. Each position predicts the next token, so the inputs $w_1, \dots, w_5$ produce predictions of $w_2, \dots, w_6$. The token w_1 is only an input and is never predicted, so it contributes

no loss term. Four tokens are predicted here: w_2, w_3, w_4, w_5 .

Step 2 — Convert each probability to a loss term.

$$-\log_2 0.5 = 1, \quad -\log_2 0.25 = 2, \quad -\log_2 0.5 = 1, \quad -\log_2 0.125 = 3.$$

Step 3 — Sum.

$$\text{NLL} = 1 + 2 + 1 + 3 = 7$$

Therefore, the correct option is **B**.

Reference: Lecture 29, Slide 4 (PDF page 92) — Pretraining Decoders (inputs $w_1, \dots, w_5$ produce predictions $w_2, \dots, w_6$)

6. A team continues pretraining a released RoBERTa checkpoint on a large corpus of appliance e-manuals. In-domain question answering improves substantially, but the model's average GLUE score falls below that of the original checkpoint. Which diagnosis and remedy apply? **[1 Mark]**

- A) Catastrophic forgetting; replace the model vocabulary
- B) Catastrophic forgetting; mix general pretraining data
- C) Overfitting to GLUE; use early stopping
- D) Curse of multilinguality; use temperature sampling

Answer & Solution

Answer: B) Catastrophic forgetting; mix general pretraining data

Solution:

In-domain performance rising while general-domain performance falls is catastrophic forgetting: continued pretraining optimises on a narrower distribution, and nothing in the objective rewards retaining behaviour on text the model no longer sees. The remedy is to keep the original distribution present in the training signal by mixing general pretraining data into the continued pretraining (or by mixing parameters, mix-out).

Therefore, the correct option is **B**.

Reference: Lecture 30, Slide 8 (PDF page 121) — Side-effect of continual pretraining: Catastrophic Forgetting

7. A 2-layer biLM produces, for a token k , the set $R_k = \{\mathbf{h}_{k,j}^{LM} \mid j = 0, 1, 2\}$, where $\mathbf{h}_{k,0}^{LM} = [\mathbf{x}_k; \mathbf{x}_k]$ is the context-independent token layer and, for $j = 1, 2$, $\mathbf{h}_{k,j}^{LM} = [\vec{\mathbf{h}}_{k,j}; \overleftarrow{\mathbf{h}}_{k,j}]$. You are given

$$\mathbf{x}_k = [1, 0], \quad \vec{\mathbf{h}}_{k,1} = [2, 1], \quad \overleftarrow{\mathbf{h}}_{k,1} = [0, 3], \quad \vec{\mathbf{h}}_{k,2} = [1, 4], \quad \overleftarrow{\mathbf{h}}_{k,2} = [2, 0].$$

For a particular task the softmax-normalised mixture weights are $s_0^{task} = 0.1$, $s_1^{task} = 0.5$, $s_2^{task} = 0.4$ and the scale factor is $\gamma^{task} = 2$. What is ELMo_k^{task} ? **[2 Marks]**

- A) [1.5, 2.1, 0.9, 1.5]
- B) [2.67, 3.33, 2.00, 2.00]
- C) [2.0, 8.0, 4.0, 0.0]
- D) [3.0, 4.2, 1.8, 3.0]

Answer & Solution

Answer: D) [3.0, 4.2, 1.8, 3.0]

Solution:

Step 1 — Form the three layer representations by concatenation.

$$\mathbf{h}_{k,0}^{LM} = [1, 0, 1, 0], \quad \mathbf{h}_{k,1}^{LM} = [2, 1, 0, 3], \quad \mathbf{h}_{k,2}^{LM} = [1, 4, 2, 0].$$

Step 2 — Compute the weighted sum $\sum_j s_j \mathbf{h}_{k,j}$.

$$\text{comp. 1: } 0.1(1) + 0.5(2) + 0.4(1) = 1.5$$

$$\text{comp. 2: } 0.1(0) + 0.5(1) + 0.4(4) = 2.1$$

$$\text{comp. 3: } 0.1(1) + 0.5(0) + 0.4(2) = 0.9$$

$$\text{comp. 4: } 0.1(0) + 0.5(3) + 0.4(0) = 1.5$$

Step 3 — Multiply by $\gamma^{task} = 2$.

$$\mathbf{ELMo}_k^{task} = 2 \cdot [1.5, 2.1, 0.9, 1.5] = [3.0, 4.2, 1.8, 3.0].$$

Therefore, the correct option is **D**.

Reference: [Lecture 26, Slide 20 \(PDF page 20\)](#) — [ELMo: More Details](#) ($\mathbf{ELMo}_k^{task} = \gamma^{task} \sum_{j=0}^L s_j^{task} \mathbf{h}_{k,j}^{LM}$)